

CALCULUS II

WEEK X

GRADIENTS AND DIRECTIONAL DERIVATIVES

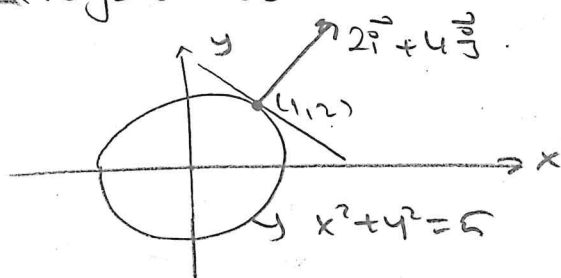
Definition At any point (x,y) where the first partial derivatives of the function $f(x,y)$ exist, we define the gradient vector

$$\nabla f(x,y) = \text{grad}(f(x,y)) \text{ by } (\nabla \text{ called del or nabla) or vector differential operator}$$
$$\nabla f(x,y) = \text{grad}(f(x,y)) = f_1(x,y)\vec{i} + f_2(x,y)\vec{j}$$

$$\nabla = \vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} \quad \nabla f(x,y) = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} \right) f(x,y) = f_1(x,y)\vec{i} + f_2(x,y)\vec{j}$$

Ex If $f(x,y) = x^2 + y^2$ then $\nabla f(x,y) = 2x\vec{i} + 2y\vec{j}$ $\nabla f(1,2) = 2\vec{i} + 4\vec{j}$

Observe that this vector is perpendicular to the tangent line $x+2y=5$ to the circle $x^2+y^2=5$ at $(1,2)$ ($1^2+2^2=5$)



Thm If $f(x,y)$ is differentiable at the point (a,b) and $\nabla f(a,b) \neq 0$ then $\nabla f(a,b)$ is a normal vector to the level curve of f that passes through (a,b)

Proof Let $r = r(t) = x(t)\vec{i} + y(t)\vec{j}$ be parametrization of the level curve of f such that $x(0)=a, y(0)=b$. Differentiating $f(x(t), y(t)) = C$ w.r.t t using Chain Rule, we obtain

$$\frac{df}{dt} = f_1(x(t), y(t)) \frac{dx}{dt} + f_2(x(t), y(t)) \frac{dy}{dt} = 0$$

For $t=0$, it says

$$\nabla f(a,b) \cdot \frac{dr}{dt} \Big|_{t=0} = 0, \text{ which means } \nabla f(a,b) \perp \frac{dr}{dt} \Big|_{t=0}$$

Def Directional Derivatives.

Let $\vec{u} = u\vec{i} + v\vec{j}$ be a unit vector so that $u^2 + v^2 = 1$.

The directional derivative of $f(x,y)$ at (a,b) in the direction of $\vec{u}$ is the rate of change of $f(x,y)$ with respect to distance measured at (a,b) along a ray in the direction of $\vec{u}$ in the xy -plane. This directional derivative is given by

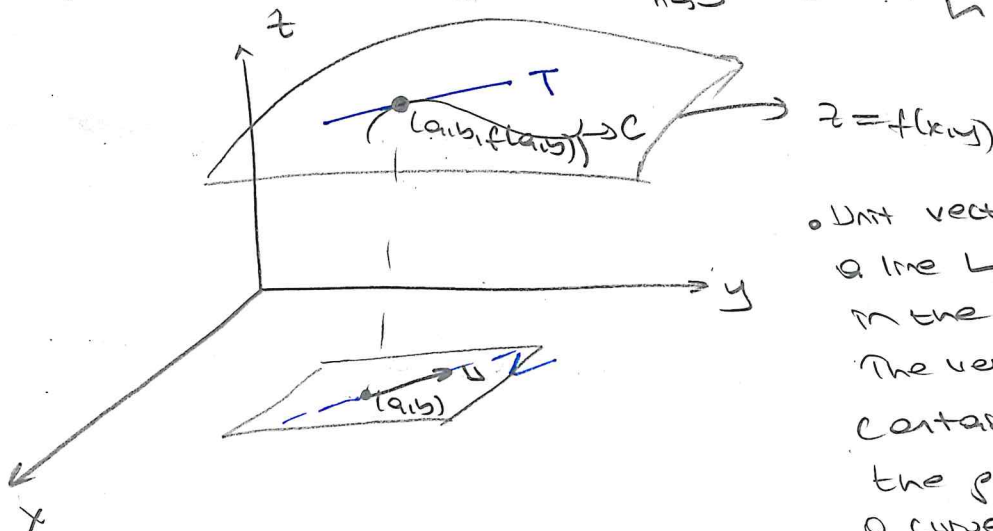
$$D_{\vec{u}} f(a,b) = \lim_{h \rightarrow 0} \frac{f(a+hu, b+hv) - f(a,b)}{h}$$

It is also given by

$$D_{\vec{u}} f(a,b) = \left. \frac{d}{dt} f(a+tu, b+tv) \right|_{t=0} \quad \text{if the derivative on the right side exists}$$

(solve Example at page 6)

Note $\vec{u} = \vec{i} = (1,0) \quad f_x(a,b) = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a,b)}{h}$
 $\vec{u} = \vec{j} = (0,1) \quad f_y(a,b) = \lim_{h \rightarrow 0} \frac{f(a, b+h) - f(a,b)}{h}$



- Unit vector $\vec{u}$ determines a line L through (a,b) in the domain of f . The vertical plane containing L intersects the graph of f in a curve C whose tangent T at $(a,b, f(a,b))$ has slope $D_{\vec{u}} f(a,b)$.

Thm (Using the gradient to find directional derivatives)

If f is differentiable at (a,b) and $\vec{u} = u\vec{i} + v\vec{j}$ is a unit vector, then the directional derivative of f at (a,b) in the direction of $\vec{u}$ is given by

$$D_{\vec{u}} f(a,b) = \vec{u} \cdot \nabla f(a,b)$$

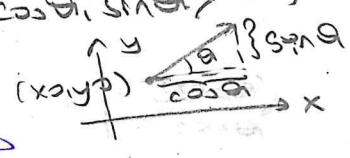
Proof By the chain rule:

$$\begin{aligned} D_{\vec{u}} f(a,b) &= \left. \frac{d}{dt} (f(a+tu, b+tv)) \right|_{t=0} \\ &= u f_1(a,b) + v f_2(a,b) = \vec{u} \cdot \nabla f(a,b) \end{aligned}$$

→ Given any nonzero vector $\vec{v}$, we can always obtain a unit vector in the same direction by dividing $\vec{v}$ by its length. The directional derivative of f at (a,b) in the direction of $\vec{v}$ is therefore given by

$$D_{\frac{\vec{v}}{|\vec{v}|}} f(a,b) = \frac{\vec{v}}{|\vec{v}|} \cdot \nabla f(a,b)$$

Remark If the unit vector $\vec{u}$ makes an angle θ with the positive x -axis, then we can write $\vec{u} = \langle \cos\theta, \sin\theta \rangle$ so, we have

$$D_{\vec{u}} f(x,y) = f_x(x,y) \cos\theta + f_y(x,y) \sin\theta$$


Ex Find the rate of change of $f(x,y) = y^4 + 2xy^3 + x^2y^2$ at $(0,1)$ measured in each of the following directions:

- (a) $\vec{i} + 2\vec{j}$ (b) $\frac{2}{3}\vec{i} - 2\vec{j}$ (c) $3\vec{i}$ (d) $\frac{1}{5}\vec{i} + \frac{4}{5}\vec{j}$

$$\nabla f = (2y^3 + 2xy^2)\vec{i} + (4y^3 + 6xy^2 + 2x^2y)\vec{j} \quad \nabla f(0,1) = 2\vec{i} + 4\vec{j}$$

(a) Unit vector in the direction of $\vec{i} + 2\vec{j}$ is $\frac{\vec{i} + 2\vec{j}}{\sqrt{5}}$. Thus, directional derivative of f at $(0,1)$ in that direction is

$$\frac{\vec{i} + 2\vec{j}}{\sqrt{5}} \cdot \nabla f(0,1) = \left(\frac{\vec{i} + 2\vec{j}}{\sqrt{5}} \right) \cdot (2\vec{i} + 4\vec{j}) = \frac{2+8}{\sqrt{5}} = 2\sqrt{5}$$

Note $\vec{i} + 2\vec{j}$ points in the same direction as $\nabla f(0,1)$ so the directional derivative is positive and equal to the length of $\nabla f(0,1)$

(b) Unit vector in the direction of $\vec{j} - 2\vec{i}$ is $\frac{\vec{j} - 2\vec{i}}{\sqrt{5}}$.

Thus, the directional derivative of f at $(0,1)$ in that direction is

$$-\frac{2\vec{i} + \vec{j}}{\sqrt{5}} \cdot (2\vec{i} + 4\vec{j}) = -\frac{4+4}{\sqrt{5}} = 0$$

Note since $\vec{j} - 2\vec{i}$ is perpendicular to $\nabla f(0,1)$, it is tangent to the level curve of f through $(0,1)$, so the directional derivative in that direction is zero.

(c) The unit vector in the direction of $3\vec{i}$ is just $\vec{i}$.

So, $\vec{i} \cdot (2\vec{i} + 4\vec{j}) = 2.$

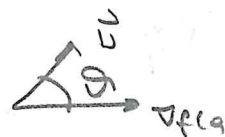
As noted previously, the directional derivative of f in the direction of the positive x -axis is just $f_x(0,1)$.

(d) Unit vector in the direction of $\vec{i} + \vec{j}$ is $\frac{\vec{i} + \vec{j}}{\sqrt{2}}$, so the directional derivative of f at $(0,1)$ in that direction is

$$\frac{\vec{i} + \vec{j}}{\sqrt{2}} \cdot (2\vec{i} + 4\vec{j}) = \frac{2+4}{\sqrt{2}} = 3\sqrt{2}$$

→ For any unit vector $\vec{u}$, we have

$$D_{\vec{u}} f(a,b) = \vec{u} \cdot \vec{\nabla} f(a,b) = |\vec{\nabla} f(a,b)| \cos \theta$$



where θ is the angle between the vectors $\vec{u}$ and $\vec{\nabla} f(a,b)$

→ Since $\cos \theta$ only takes values between -1 and 1 , $D_{\vec{u}} f(a,b)$ only takes values between:

$$|\nabla f(a,b)| \leq D_{\vec{u}} f(a,b) \leq -|\nabla f(a,b)|$$

$$D_{\vec{u}} f(a,b) = -|\nabla f(a,b)|$$



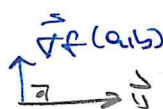
• $\vec{u}$ and $\vec{\nabla} f(a,b)$
different direction
• $\cos \pi = -1$

$$D_{\vec{u}} f(a,b) = |\nabla f(a,b)|$$



• $\vec{u}$ and $\vec{\nabla} f(a,b)$
same direction
• $\cos 0 = 1$

$$D_{\vec{u}} f(a,b) = 0$$



• at the direction of the
(tangent line to the)
level curve of f
through (a,b)

Geometric Properties of the gradient vector

(i) At (a,b) , $f(x,y)$ increases most rapidly in the direction of the gradient vector $\vec{\nabla} f(a,b)$. The maximum rate of increase is $|\vec{\nabla} f(a,b)|$

(ii) At (a,b) , $f(x,y)$ decreases most rapidly in the direction of $-\vec{\nabla} f(a,b)$. The maximum rate of decrease is $-|\vec{\nabla} f(a,b)|$

(iii) The rate of change of $f(x,y)$ at (a,b) is zero in direction tangent to the level curve of f that passes through (a,b)

Ex Find the derivative of $f(x,y) = x^2 + xy$ at $P_0(1,2)$ in the direction of the unit vector $\vec{u} = \frac{1}{\sqrt{2}}\vec{i} + \frac{1}{\sqrt{2}}\vec{j}$

(a) Use formal definition

$$\begin{aligned} D_{\vec{u}}f(P_0) &= \left. \frac{df}{ds} \right|_{P_0} = \lim_{h \rightarrow 0} \frac{f(x_0 + h\vec{u}, y_0 + h\vec{v}) - f(x_0, y_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{\sqrt{2}}, 2 + \frac{h}{\sqrt{2}}\right) - f(1,2)}{h} = \lim_{h \rightarrow 0} \frac{\left(1 + \frac{h}{\sqrt{2}}\right)^2 + \left(1 + \frac{h}{\sqrt{2}}\right)\left(2 + \frac{h}{\sqrt{2}}\right) - 3}{h} \\ &= \lim_{h \rightarrow 0} \frac{\left(1 + \frac{2h}{\sqrt{2}} + \frac{h^2}{2}\right) + \left(2 + \frac{3h}{\sqrt{2}} + \frac{h^2}{2}\right) - 3}{h} \\ &= \lim_{h \rightarrow 0} \frac{5\frac{h}{\sqrt{2}} + h^2}{h} = \lim_{h \rightarrow 0} \left(\frac{5}{\sqrt{2}} + h\right) = \frac{5}{\sqrt{2}} \end{aligned}$$

(b) Use grad vector

$$\left. \nabla f \right|_{(1,2)} = \langle 2x+y, x \rangle \Big|_{(1,2)} = \langle 4, 1 \rangle$$

$$D_{\vec{u}}f(P_0) = \langle 4, 1 \rangle \cdot \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle = 5/\sqrt{2}$$

Ex Find directions of $f(x,y) = \frac{x^2}{2} + \frac{y^2}{2}$ in which

(a) f increases most rapidly at point $(1,1)$

(b) f decreases most rapidly at point $(1,1)$

(c) What are the direction of zero change of f at $(1,1)$

$$\begin{aligned} \text{(a)} \quad \left. (\nabla f) \right|_{(1,1)} &= \left(x\vec{i} + y\vec{j} \right) \Big|_{(1,1)} = \vec{i} + \vec{j} \quad \text{direction is } \vec{D} = \frac{\nabla f(1,1)}{|\nabla f(1,1)|} \\ \vec{D} &= \frac{\vec{i} + \vec{j}}{\sqrt{2}} \quad \text{increases most rapidly} \end{aligned}$$

$$\text{(b)} \quad \text{decreases most rapidly} \quad -\vec{D} = -\frac{1}{\sqrt{2}}\vec{i} - \frac{1}{\sqrt{2}}\vec{j}$$

(c)

© zero change at $(1,1)$ are the directions orthogonal to $\vec{\nabla} f$ $\vec{u} = \langle u_1, u_2 \rangle$ such that $\vec{\nabla} f \cdot \vec{u} = 0$

$$\vec{\nabla} f|_{(1,1)} \cdot \vec{u} = \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle \cdot \langle u_1, u_2 \rangle = \frac{u_1}{\sqrt{2}} + \frac{u_2}{\sqrt{2}} = 0 \quad u_1 = -u_2$$

$\vec{u}$ must be a unit vector so $\sqrt{u_1^2 + u_2^2} = 1$

$$\text{Since } u_2 = -u_1 \Rightarrow \sqrt{(u_1)^2 + (-u_1)^2} = 1 \quad u_1 = \pm \frac{1}{\sqrt{2}}$$

So, zero change at the directions orthogonal to $\vec{\nabla} f$ at $(1,1)$:

$$\vec{n} = -\frac{1}{\sqrt{2}}\vec{i} + \frac{1}{\sqrt{2}}\vec{j} \quad \text{and} \quad -\vec{n} = \frac{1}{\sqrt{2}}\vec{i} - \frac{1}{\sqrt{2}}\vec{j}$$

Ex 7 The temperature at position (x,y) in a region of the xy -plane is $T^\circ\text{C}$ where

$$T(x,y) = x^2 e^{-y}$$

Find in what direction at the point $(2,1)$ does the temperature increase most rapidly? What is the rate of increase of f in that direction

$$\vec{\nabla} T(x,y) = 2x e^{-y} \vec{i} - x^2 e^{-y} \vec{j}$$

$$\vec{\nabla} T(2,1) = \frac{4}{e} \vec{i} - \frac{4}{e} \vec{j} = \frac{4}{e} (\vec{i} - \vec{j})$$

• $T(x,y)$ increases most rapidly in the direction of $\frac{4}{e} (\vec{i} - \vec{j})$ (we can say also $\vec{i} - \vec{j}$). If we obtain unit vector

$$\frac{\vec{u}}{|\vec{u}|} = \frac{\frac{4}{e} (\vec{i} - \vec{j})}{\frac{4}{e} \sqrt{2}} = \frac{\vec{i} - \vec{j}}{\sqrt{2}} \quad |\vec{u}| = \sqrt{\frac{16}{e^2} + \frac{16}{e^2}} = \sqrt{\frac{32}{e^2}} = \frac{4}{e} \sqrt{2}$$

• The rate of increase in this direction is

$$|\vec{\nabla} T(2,1)| = 4\sqrt{2}/e \text{ } ^\circ\text{C/unit distance.}$$

Ex Suppose that the height of a hill above sea level is given by $z = 1000 - 0.01x^2 - 0.02y^2$. If you are at the point $(60, 100)$ in what direction is the elevation changing fastest? What is the maximum rate of change of the elevation at this point.

$$\text{grad } f = \nabla f = \langle -0.02x, -0.04y \rangle$$

$$\text{grad } f|_{(60, 100)} = \langle -1.2, -4 \rangle$$

Maximum rate of change $|\text{grad } f|_{(60, 100)}| = \sqrt{(-1.2)^2 + (-4)^2} = \sqrt{17.2}$

Ex A hiker is standing beside a stream on the side of a mountain. The height of land (in metres) at any point (x, y) is given by the function

$$h(x, y) = \frac{20,000}{3 + x^2 + 2y^2}$$

hiker point is $(3, 2)$



(a) What is the direction of the flow of the stream at $(3, 2)$ on the hiker's map? How fast is the stream descending at her location?

(b) Find the equation of the path of the stream on the hiker's map.

(c) At what angle the path of the stream (on the map) should the hiker set out if she wishes to climb at 15° inclination to the horizontal?

(d) Make a sketch of the hiker's map, showing some curves of constant elevation, and showing stream.

② $\nabla h(x,y) = - \frac{20.000}{(3+x^2+2y^2)^2} (2x\vec{i} + 4y\vec{j})$

$\nabla h(3,2) = -100 (3\vec{i} + 4\vec{j})$

$|\nabla h(3,2)| = 500 \rightarrow$ The stream is flowing in the direction whose horizontal projection at $(3,2)$ is $-\nabla h(3,2)$, that is, in the horizontal direction $3\vec{i} + 4\vec{j}$.

The stream is descending at a rate of 500 m/km , that is, 0.5 m per horizontal meter travelled

③ If the vector $dr = dx\vec{i} + dy\vec{j}$ is tangent to the path of the stream at point (x,y) on the map, then dr is parallel to $\nabla h(x,y)$. Hence, the components of these two vectors are parallel.

$$\frac{dx}{2x} = \frac{dy}{4y} \quad \text{or} \quad \frac{dy}{y} = \frac{2dx}{x}$$

Integrating both sides gives $\ln y = 2 \ln x + \ln C$ or $y = Cx^2$

Since path of the stream passes through $(3,2)$, we have

$C = 2/9$ and equation is $y = 2x^2/9$

④ Suppose hiker moves away from $(3,2)$ in the direction of the unit vector $\vec{u}$. She will be ascending at an angle

of 15° if the directional derivative of h

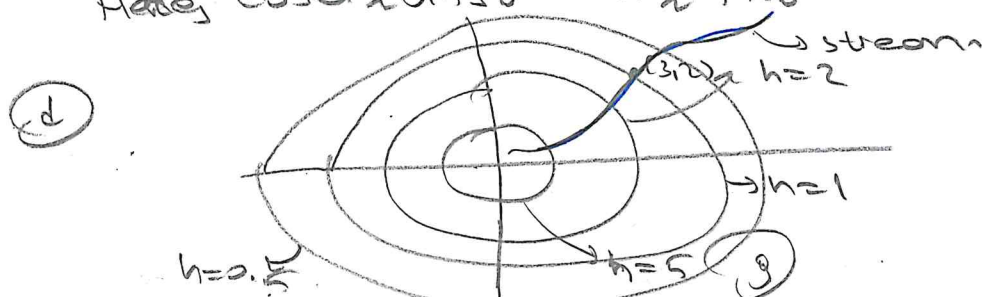
in the direction of $\vec{u}$ is $1000 \times 15^\circ \approx 268$ (The 1000

compensates for the fact that the vertical units are

meters while horizontal are kilometers). If θ is the angle between $\vec{u}$ and the upstream direction, then

$$500 \cos \theta = |\nabla h(3,2)| \cos \theta = D_{\vec{u}} h(3,2) \approx 268$$

Hence, $\cos \theta \approx 0.536$ $\theta \approx 57.6^\circ$



The Gradient in Three and More Dimensions

→ By analogy with two-dimensional case, a function $f(x_1, x_2, \dots, x_n)$ of n variables possessing first partial derivatives has gradient given by

$$\nabla f(x_1, x_2, \dots, x_n) = \frac{\partial f}{\partial x_1} \vec{e}_1 + \frac{\partial f}{\partial x_2} \vec{e}_2 + \dots + \frac{\partial f}{\partial x_n} \vec{e}_n$$

where $\vec{e}_j$ is the unit vector from the origin to the unit point on the j th coordinate axes. In particular, for a function of three variables

$$\nabla f(x, y, z) = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$$

Ex Let $f(x, y, z) = x^2 + y^2 + z^2$

(a) Find $\nabla f(1, -1, 2) = ?$ $\nabla f = (2x, 2y, 2z) \Big|_{(1, -1, 2)} = (2, -2, 4)$

(b) Find equation of the tangent plane $x^2 + y^2 + z^2 = 6$ at point $(1, -1, 2)$

$\nabla f(1, -1, 2) = (2, -2, 4)$ is normal of the plane

Equation of the plane $2(x-1) - 2(y+1) + 4(z-2) = 0$

(c) The maximum increase of f at f ? It is $|\nabla f(1, -1, 2)| = 2\sqrt{6}$

at direction $\vec{u} = \frac{\nabla f(1, -1, 2)}{|\nabla f(1, -1, 2)|} = \frac{(2, -2, 4)}{\sqrt{4+4+16}} = \frac{(2, -2, 4)}{2\sqrt{6}} = \frac{(1, -1, 2)}{\sqrt{6}}$

(d) What is the rate of change of f w.r.t distance of f at $(1, -1, 2)$ measured in the direction from that point towards the point $(3, 1, 1)$?

The direction from $(1, -1, 2)$ towards $(3, 1, 1)$ is specified by $2\vec{i} + 2\vec{j} - \vec{k}$. The rate of change of f w.r.t distance in this direction is

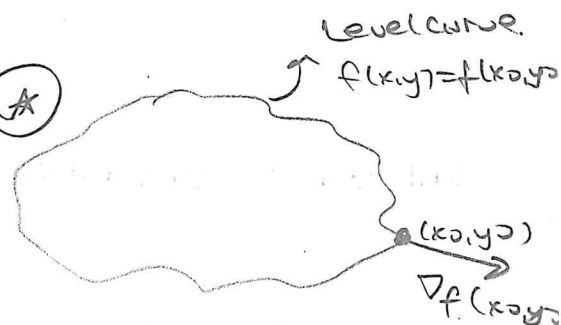
$$\frac{2\vec{i} + 2\vec{j} - \vec{k}}{\sqrt{4+4+1}} \cdot (2\vec{i} - 2\vec{j} + 4\vec{k}) = \frac{4-4-4}{3} = -\frac{4}{3}$$

GRADIENTS AND TANGENTS TO LEVEL CURVES

→ If a differentiable function $f(x,y)$ has a constant value c along a smooth curve $\vec{r} = g(t)\vec{i} + h(t)\vec{j}$, (making the curve a level curve of f), then $f(g(t), h(t)) = c$. Differentiating both sides of this equation w.r.t. t leads to equation

$$\frac{d}{dt} f(g(t), h(t)) = \frac{d}{dt}(c) \Rightarrow \underbrace{\frac{\partial f}{\partial x} \frac{dg}{dt} + \frac{\partial f}{\partial y} \frac{dh}{dt}}_{\text{Chain Rule}} = 0$$

$$\underbrace{\left(\frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} \right)}_{\vec{\nabla} f} \cdot \underbrace{\left(\frac{dg}{dt} \vec{i} + \frac{dh}{dt} \vec{j} \right)}_{\frac{d\vec{r}}{dt}} = 0 \quad (*)$$



Eq. (*) says that $\vec{\nabla} f$ is normal to the tangent vector

$\frac{d\vec{r}}{dt}$, so it is normal to the curve.

(*) At every point (x_0, y_0) in the domain of a differentiable function $f(x,y)$, the gradient of f is normal to the level curve through (x_0, y_0) .

→ This observation enables us to find equations for tangent lines to level curves

→ They are the lines normal to the gradients. The line through a point $P_0(x_0, y_0)$ normal to a vector $\vec{N} = A\vec{i} + B\vec{j}$ has the equation

$$A(x - x_0) + B(y - y_0) = 0$$

→ If $\vec{N}$ is the gradient $(\vec{\nabla} f)_{(x_0, y_0)} = f_x(x_0, y_0)\vec{i} + f_y(x_0, y_0)\vec{j}$, the equation is the tangent line given by

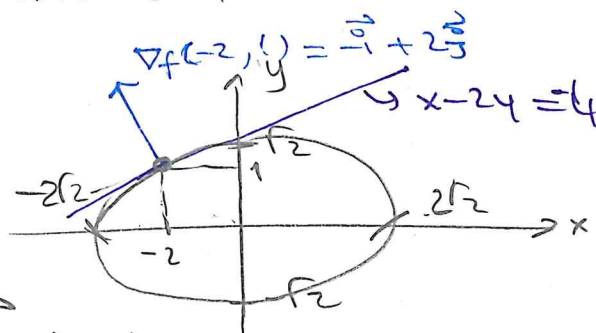
$$\boxed{f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) = 0}$$

Ex Find an equation for the tangent to the ellipse.

$$\frac{x^2}{4} + y^2 = 2 \quad \text{at point } (-2, 1)$$

The ellipse is the level curve

$$\text{of } f(x, y) = \frac{x^2}{4} + y^2$$



The gradient of f at $(-2, 1)$ is

$$\nabla f|_{(-2, 1)} = \left(\frac{x}{2}\vec{i} + 2y\vec{j} \right)|_{(-2, 1)} = -\vec{i} + 2\vec{j}$$

The tangent line is $(-1)(x+2) + 2(y-1) = 0 \Rightarrow x - 2y = -4$

TANGENT PLANES AND NORMAL LINES

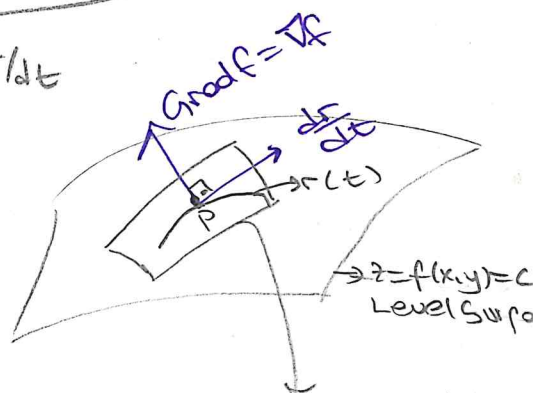
→ If $\vec{r}(t) = g(t)\vec{i} + h(t)\vec{j} + k(t)\vec{k}$ is a smooth curve on the level surface $f(x, y, z) = c$ of a differentiable function f , then $f(g(t), h(t), k(t)) = c$.

→ Differentiating both sides w.r.t. t leads to.

$$\frac{d}{dt} f(g(t), h(t), k(t)) = \frac{d}{dt}(c) \xrightarrow[\text{Rule}]{\text{Chain}} \frac{\partial f}{\partial x} \frac{dg}{dt} + \frac{\partial f}{\partial y} \frac{dh}{dt} + \frac{\partial f}{\partial z} \frac{dk}{dt} = 0$$

$$\underbrace{\left(\frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j} + \frac{\partial f}{\partial z}\vec{k} \right)}_{\nabla f} \cdot \underbrace{\left(\frac{dg}{dt}\vec{i} + \frac{dh}{dt}\vec{j} + \frac{dk}{dt}\vec{k} \right)}_{d\vec{r}/dt} = 0$$

→ At every point along the curve, ∇f is orthogonal to the curve's velocity vectors.



****** The velocity vectors at P all lie in a common plane, which we call tangent plane to the level surface at P .

Tangent plane to level curve Let $P = (x_0, y_0, z_0)$ If $\nabla f(x_0, y_0, z_0) \neq 0$, it is therefore natural to define tangent plane to the level surface

$f(x_0, y_0, z_0) = c$ at $P(x_0, y_0, z_0)$ as the plane passes through P and has a normal vector $\nabla f(x_0, y_0, z_0)$.

Definition The tangent plane at $P_0(x_0, y_0, z_0)$ on the level surface $f(x, y, z) = c$ of a differentiable function f is the plane P_0 normal to $\nabla f|_{P_0}$.

The normal line of the surface at P_0 is the line through P_0 parallel to $\nabla f|_{P_0}$.

Tangent Plane to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$

$$f_x(P_0)(x-x_0) + f_y(P_0)(y-y_0) + f_z(P_0)(z-z_0) = 0$$

$$(\nabla f|_{P_0} \cdot \langle (x-x_0), (y-y_0), (z-z_0) \rangle = 0)$$

Normal Line to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$

$$x = x_0 + f_x(P_0)t, \quad y = y_0 + f_y(P_0)t, \quad z = z_0 + f_z(P_0)t$$

★ For special case $z = F(x, y) \Rightarrow \frac{F(x, y) - z}{f(x, y, z)} = 0 \quad f_z = -1$

Ex 7 Find the equation of the tangent plane to the surface $x^2 - y^2 + z^2 = 1$ at the point $P(2, 2, 1)$

$$f(x, y, z) = x^2 - y^2 + z^2$$

$$x^2 - y^2 + z^2 = 1 \quad \text{level surface of } f(x, y, z) = 1$$

$$\nabla f = \langle 2x, -2y, 2z \rangle$$

$$\nabla f|_P = \langle 4, -4, 2 \rangle$$

(Solving $f(x, y, z) - 1 = 0$ can be more practical for students while taking derivative. Otherwise they forget $z = F(x, y)$ is on the left side :))

$$\nabla f|_P \cdot \langle x-2, y-2, z-1 \rangle = 0 \quad \underbrace{2(x-2) - 4(y-2) + 2(z-1) = 0}_{\text{Tangent plane}}$$

Normal line $\frac{x-2}{4} = \frac{y-2}{-4} = \frac{z-1}{2}$ or we can

represent it in the form $x = 2 + 4t \quad y = -4t + 2$
 $z = 2t + 1$

Ex 7 Find a unit normal vector $\vec{u}$ of $z^2 = y^2 - x^2$ at the point $(2, 2, 1)$

$$f(x, y, z) = z^2 - y^2 + x^2 \quad f(2, 2, 1) = 1$$

So, the point $(2, 2, 1)$ is on the level surface S given by

$$f(x, y, z) = z^2 - y^2 + x^2 = 1 \quad \nabla f \text{ is a normal vector of } S$$

at point $(2, 2, 1)$ find $\nabla f(2, 2, 1) = \langle 4, -4, 2 \rangle$

Unit vector $\frac{\nabla f}{|\nabla f|} = \langle \frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \rangle$

Ex 7 Find the tangent plane and normal line of the surface

$$f(x,y,z) = x^2 + y^2 + z - 9 = 0$$

at point $P(1,2,4)$

$$\nabla f = \langle 2x, 2y, 1 \rangle$$

$$\text{Tangent plane } 2(x-1) + 4(y-2) + 1(z-4) = 0$$

$$\nabla f|_{(1,2,4)} = \langle 2, 4, 1 \rangle$$

$$\text{Normal line } \frac{x-1}{2} = \frac{y-2}{4} = \frac{z-4}{1}$$

Ex Find the tangent plane to the surface

$$z = x \cos y - y e^x \text{ at } (0,0,0)$$

$$f(x,y,z) = x \cos y - y e^x - z$$

$$\nabla f = \langle \cos y - y e^x, -x \sin y - e^x, -1 \rangle$$

$$\nabla f|_{(0,0,0)} = \langle \cos 0 - 0, 0 - e^0, -1 \rangle = \langle 1, -1, -1 \rangle$$

$$\text{Tangent surface } 1(x-0) - 1(y-0) - 1(z-0) = 0$$

$$x - y - z = 0$$

$$\text{Normal line } \frac{x-0}{1} = \frac{y-0}{-1} = \frac{z-0}{-1}$$

Ex Find the tangent plane and normal line to the surface $f(x,y,z) = x^2 + y^2 + z - 9 = 0$ at point

$P_0(1,2,4)$

$$\nabla f|_{P_0} = \langle 2x, 2y, 1 \rangle|_{P_0} = \langle 2, 4, 1 \rangle$$

$$\text{Tangent plane } 2(x-1) + 4(y-2) + 1(z-4) = 0$$

$$\text{Normal line } \frac{x-1}{2} = \frac{y-2}{4} = \frac{z-4}{1}$$

$$x = 1 + 2t \quad y = 2 + 4t \quad z = 4 + t$$

→ Tangent plane to a smooth curve $z = f(x, y)$ at a point $P_0(x_0, y_0, f(x_0, y_0))$ is the tangent plane through P_0 to the zero level surface of the function $F(x, y, z) = f(x, y) - z$

$$\frac{\partial F}{\partial x} = \frac{\partial f}{\partial x}, \quad \frac{\partial F}{\partial y} = \frac{\partial f}{\partial y}, \quad \frac{\partial F}{\partial z} = -1$$

So, the plane equation

$$\frac{\partial f}{\partial x} \bigg|_{P_0} (x - x_0) + \frac{\partial f}{\partial y} \bigg|_{P_0} (y - y_0) - (z - z_0) = 0 \rightarrow \text{Tangent plane.}$$

$$\frac{x - x_0}{f_x(P_0)} = \frac{y - y_0}{f_y(P_0)} = \frac{z - z_0}{-1} \rightarrow \text{Normal line.}$$

Ex 7 Find the plane tangent to the surface $z = x \cos y - y e^x$ at $(0, 0, 0)$.

$$f(x, y, z) = x \cos y - y e^x - z \quad z|_{(0,0)} = 0 \cdot \cos 0 - 0 \cdot e^0 = 0$$

$$\nabla f = \langle \cos y - y e^x, -x \sin y - e^x, -1 \rangle$$

$$\nabla f|_{(0,0,0)} = \langle \cos 0 - 0, 0 - e^0, -1 \rangle = \langle 1, -1, -1 \rangle$$

$$1(x-0) - 1(y-0) - 1(z-0) = 0 \quad \text{Tangent plane}$$

$$x - y - z = 0$$

$$\frac{x-0}{1} = \frac{y-0}{-1} = \frac{z-0}{-1} \quad \text{Normal line.}$$

Ex 7 The surfaces $f(x, y, z) = x^2 + y^2 - z = 0$ (A cylinder) and $g(x, y, z) = x + z - 4 = 0$ (A plane) meet in an ellipse E . Find parametric equations for the tangent line to E at point $P_0(1, 1, 3)$.

$$\nabla f|_{(1,1,3)} = \langle 2x, 2y, -1 \rangle = \langle 2, 2, -1 \rangle$$

$$\nabla g|_{(1,1,3)} = \langle 1, 0, 1 \rangle$$

$$\eta = \nabla f|_{(1,1,3)} \times \nabla g|_{(1,1,3)} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & -1 \\ 1 & 0 & 1 \end{vmatrix}$$

Then, tangent line orthogonal both $\nabla f, \nabla g$. $\eta = \langle 2, -2, -2 \rangle = \langle 1, -1, -1 \rangle$

$$\frac{x-1}{2} = \frac{y-1}{-2} = \frac{z-3}{-2} \quad (15)$$

$$\eta \perp \nabla f$$

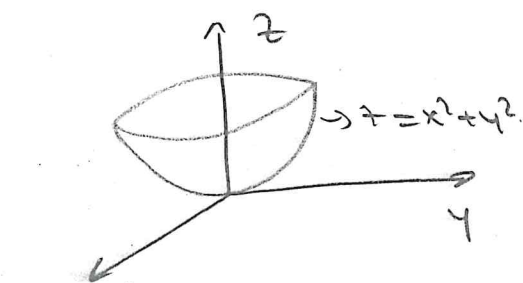
$$\eta \perp \nabla g$$

APPLICATIONS OF PARTIAL DERIVATIVES

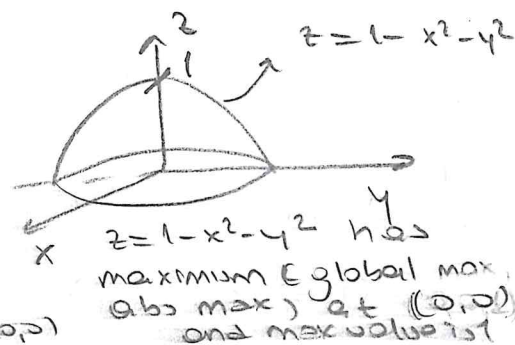
MAX-MIN PROBLEM

- $f(x,y)$ has a **local maximum** at (a,b) if $f(x,y) \leq f(a,b)$ when (x,y) is near (a,b) local max value
- $f(x,y)$ has a **local minimum** at (a,b) if $f(x,y) \geq f(a,b)$ when (x,y) is near (a,b) local min. value

If the inequalities hold for ALL points (x,y) in the domain of f , then f has an **absolute maximum** (or **absolute minimum**) at (a,b)



$z = x^2 + y^2$ has minimum at origin $(0,0)$
(global minimum, absolute minimum)
and min value is 0



$z = 1 - x^2 - y^2$ has maximum (global max, abs max) at $(0,0)$ and max value is 1

Theorem (Necessary conditions for extreme values)

A function $f(x,y)$ can have a local or absolute extreme value at point (a,b) in its domain only if (a,b) is one of the following:

- a **critical point** of f , that is, a point satisfying $\nabla f(a,b) = 0$
- a **singular point** of f , that is, a point $\nabla f(a,b)$ doesn't exist
- a **boundary point** of the domain f .

A set in $\mathbb{R}^n$ is **bounded** if it is contained inside some ball $x_1^2 + x_2^2 + \dots + x_n^2 \leq R^2$ of finite radius R .

A set on the real line is **bounded** if it is contained in an interval of finite length.

Thm Sufficient Condition for extreme values

If f is a continuous function of n variables whose domain is a closed and bounded set in $\mathbb{R}^n$, then the range of f is bounded set of real numbers, and there are points in its domain where f takes on absolute maximum and minimum values.

Ex $f(x,y) = x^2 + y^2 - 2x - 6y + 14$ find abs. max, min points.

$$\begin{aligned} f_x(x,y) &= 2x - 2 \Rightarrow x = 1 \\ f_y(x,y) &= 2y - 6 \Rightarrow y = 3 \end{aligned} \quad \left. \vphantom{\begin{aligned} f_x(x,y) &= 2x - 2 \\ f_y(x,y) &= 2y - 6 \end{aligned}} \right\} \begin{aligned} f_x(x,y) &= 0 \\ f_y(x,y) &= 0 \end{aligned} \quad (1,3) \text{ critical point.}$$

Singular point and boundary point doesn't exist.

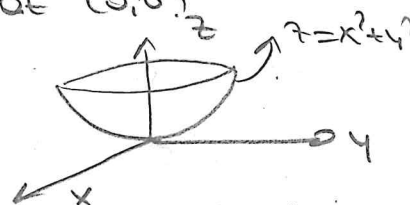
$$f(x,y) = x^2 - 2x + y^2 - 6y + 14 = (x-1)^2 + (y-3)^2 + 4$$

Since $(x-1)^2 \geq 0$ and $(y-3)^2 \geq 0$, we have $f(x,y) \geq 4$ for all (x,y) values. Then, $f(1,3) = 4$ is a local minimum and in fact it is absolute minimum of f .

Ex $f(x,y) = x^2 + y^2$

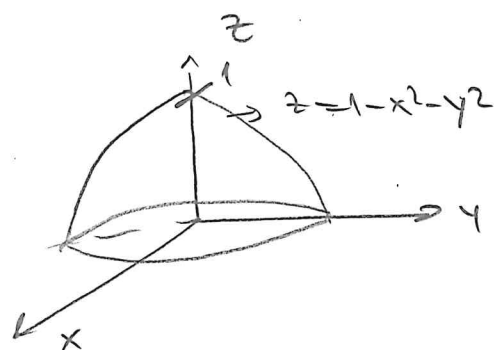
The function only have critical point at $(0,0)$

$$\begin{aligned} f_x &= 2x \Rightarrow f_x = 0 & 2x &= 0 \\ f_y &= 2y \Rightarrow f_y = 0 & 2y &= 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} f_x &= 2x \\ f_y &= 2y \end{aligned}} \right\} (0,0)$$



f has (absolute) minimum value 0 at that point. If the domain of f is not restricted f has no maximum value.

— $g(x,y) = 1 - x^2 - y^2$ has (absolute) maximum value 1 at its critical point $(0,0)$

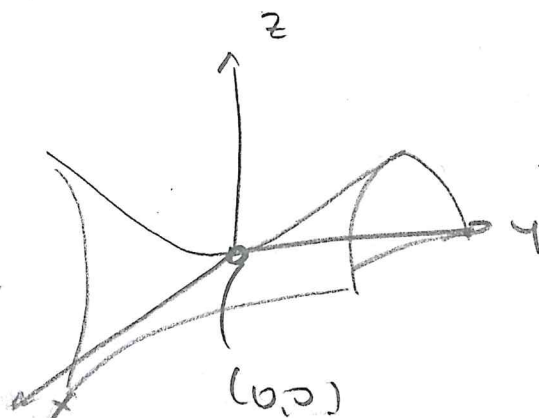


Ex 7 The function $h(x,y) = y^2 - x^2$ also has a critical point at $(0,0)$ but has neither a local maximum nor a local minimum value at that point.

$$\begin{cases} h_x = -2x \\ h_y = 2y \end{cases} \quad \begin{cases} h_x(x,y) = 0 \Rightarrow x=0 \\ h_y(x,y) = 0 \Rightarrow y=0 \end{cases} \quad \text{only critical point is } (0,0)$$

Notice that on the x -axis, we have $y=0$ so $f(x,y) = -x^2 \leq 0$
 (If $x \neq 0$), On the y -axis, we have $x=0$ so $f(x,y) = y^2 \geq 0$
 (If $y \neq 0$)

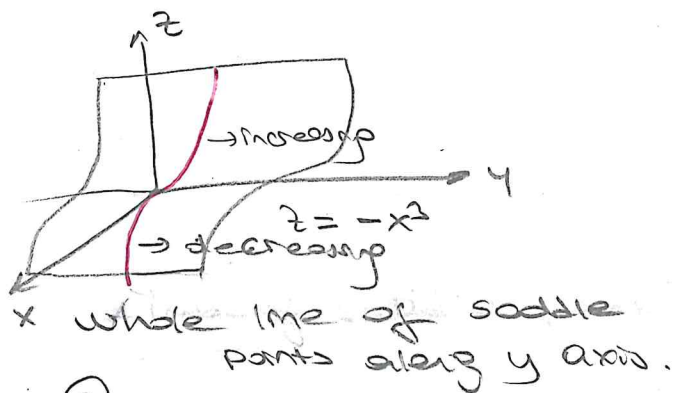
Hence, every disk with center $(0,0)$ contains where f takes positive and also negative values. Therefore, $f(0,0) = 0$ can't be an extreme value for f , so f has no extreme value



In view of the shape of the surface, we call the critical point $(0,0)$ a saddle point of h .

→ In general, we will somewhat loosely call any interior critical point of the domain of a function f of several variables a saddle point if f does not have a local maximum or minimum value there.

→ Near a saddle point, the graph will not always look like a saddle



Ex 7 The function $f(x,y) = \sqrt{x^2+y^2}$ has no critical points.

$$f_x = \frac{x}{\sqrt{x^2+y^2}}$$

$$f_x \neq 0$$

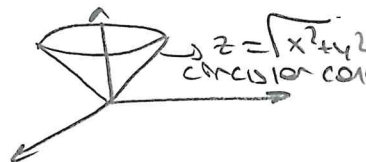
But does have a singular point at $(0,0)$

$$f_y = \frac{y}{\sqrt{x^2+y^2}}$$

$$f_y \neq 0$$

$\nabla f = (f_x, f_y)$ doesn't exist at $(0,0)$.

where the function $f(x,y)$ has a local (and absolute) minimum value which is $f(0,0) = 0$.



Ex 7 The function $f(x,y) = 1-x$ is defined everywhere in the xy plane and has no critical or singular points. $\nabla f(x,y) = -\vec{i}$. Therefore f has no extreme values.

However, if we restrict the domain of f to the points in the disk $x^2+y^2 \leq 1$ (a closed bounded set in xy plane) then f has absolute maximum and minimum values.

The maximum value is 2 at the boundary point $(-1,0)$, and the minimum value is 0 at $(1,0)$.

Ex 7 Find and classify the critical points of $f(x,y) = 2x^3 - 6xy + 3y^2$.

$$f_1(x,y) = 6x^2 - 6y \quad (\Rightarrow) \quad x^2 = y$$

$$f_2(x,y) = -6x + 6y \quad (\Rightarrow) \quad x = y$$

Together, these equations imply that $x^2 = x$ so that $x=0$ or $x=1$. Therefore, the critical points are $(0,0)$ and $(1,1)$.

Consider $(0,0)$. Here $\Delta f = f(h,k) - f(0,0)$ (h,k a point near $(0,0)$)

$$\Delta f = f(h,k) - f(0,0) = 2h^3 - 6hk + 3k^2$$

$f(h,k) - f(0,0) = 2h^3$ is positive for positive h and negative for small h , f can't have a maximum or minimum value at $(0,0)$. Therefore $(0,0)$ is a saddle point.

Now consider $(1,1)$. Here Δf is given by

$$\begin{aligned}\Delta f &= f(1+h, 1+k) - f(1,1) \\&= 2(1+h)^3 - 6(1+h)(1+k) + 3(1+k)^2 - (-1) \\&= 6h^2 - 6hk + 3k^2 + 2h^3 \\&= 3(h-k)^2 + h^2(3+2h)\end{aligned}$$

Both terms in the latter expression are nonnegative

If $|h| < 3/2$, and they are not both zero unless $h=k=0$.

Hence, $\Delta f > 0$ for small h and k , and f has local minimum value $-1 = f(1,1)$ at point $(1,1)$